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From arrest to recovery: the 2025 ERC guidelines redefine the chain of survival

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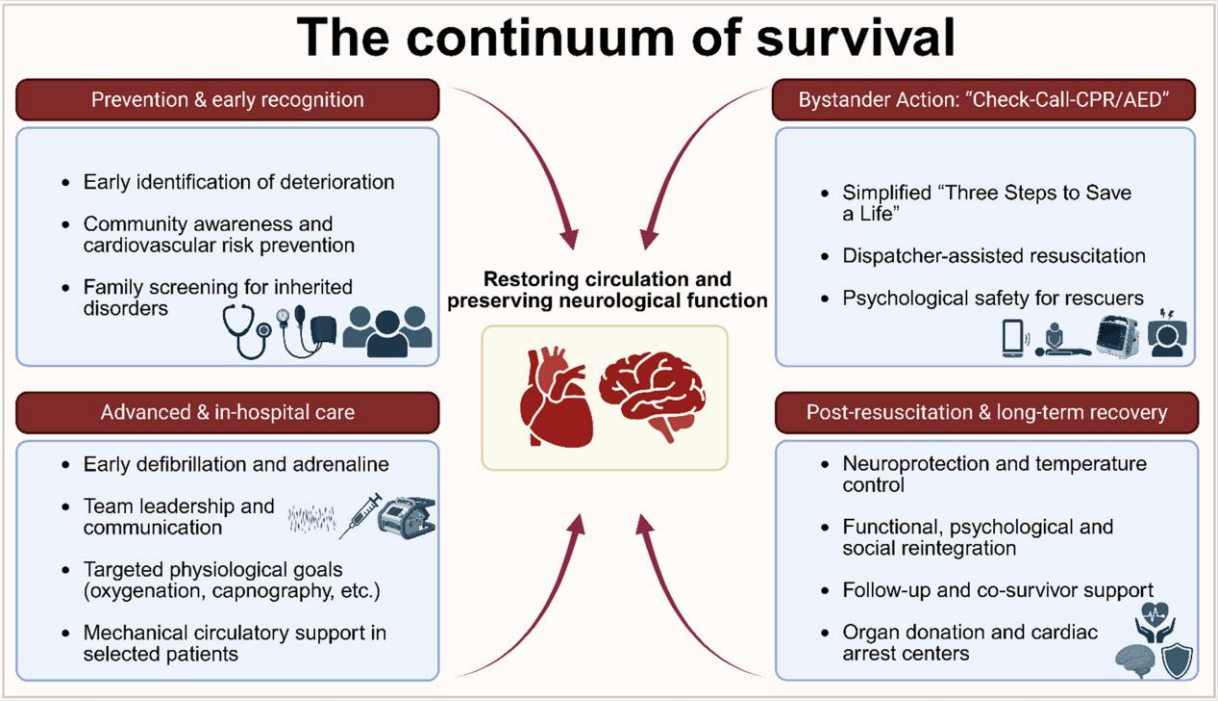
Graphical abstract



The illustration depicts the reconceptualized *Chain of Survival* as a continuous pathway from prevention to long-term recovery. It begins with prevention and early recognition, emphasizing detection of clinical deterioration, community-based cardiovascular risk reduction and family screening for inherited cardiac disorders. Bystander action ('Check-Call-CPR/AED') highlights the simplified 'Three Steps to Save a Life,' dispatcher-assisted resuscitation and psychological support for rescuers. Advanced and in-hospital care focus on rapid defibrillation and adrenaline administration, structured team communication, targeted physiological goals and selective use of mechanical circulatory support. The continuum extends to post-resuscitation and long-term recovery, encompassing multimodal neuroprotection, temperature management, rehabilitation, structured

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follow-up with co-survivor support and integration of organ donation and cardiac arrest centre networks. At its core, the *Chain of Survival* aims to restore circulation and preserve neurological function to achieve meaningful survival and quality of life. AED, automated external defibrillator; CPR, cardiopulmonary resuscitation. Created with BioRender. https://BioRender.com/sw89cbw.

The 2025 European Resuscitation Council (ERC) Guidelines mark a pivotal update in resuscitation.[1] Moving beyond a purely emergency-focused model, the recommendations emphasize continuity from early risk detection and rapid intervention to advanced life support (ALS) and structured rehabilitation. The *Chain of Survival* is reframed as a continuous pathway of preventive, therapeutic and restorative care aimed at improving neurological outcomes and long-term quality of life after cardiac arrest.

Professor Jerry P. Nolan (senior author, ERC Guidelines 2025):

> *'The 2025 ERC Guidelines are underpinned by the science reviews undertaken by the International Liaison Committee on Resuscitation and, as such, are the most evidence-based to date. We have taken a holistic approach with recommendations aimed at improving patient-centered long-term outcomes. This is reinforced by the strong focus on rehabilitation and survivorship to restore quality of life.'*

Chain of survival

While the familiar four-ring structure remains, the components have been redefined. Prevention is now explicitly embedded in the first ring, reflecting the growing recognition that early identification of deterioration saves lives. Early cardiopulmonary resuscitation (CPR) and defibrillation are consolidated into a unified second ring, highlighting their synergistic impact. Advanced and post-resuscitation care is integrated into a single continuum, and the final ring now focuses on long-term recovery, rehabilitation and reintegration into society. This reconceptualization marks a major shift from the 2021 guidelines, extending responsibility beyond hospital discharge.

Adult basic life support

The 2025 BLS recommendations prioritize simplicity, speed and accessibility through the 'Three Steps to Save a Life': 'Check, Call, CPR/AED'. Dispatcher-assisted resuscitation is strengthened, with dispatchers guiding CPR, automated external defibrillators (AED) use and first-responder activation. AED use is explicitly encouraged for all bystanders, trained or not, and psychological support for rescuers is acknowledged for the first time.

Dr. Alessandro Galluzzo (consultant cardiologist in acute and intensive cardiac care medicine):

> *'The updated Basic Life Support recommendations make lifesaving actions more intuitive and achievable for everyone. The strengthened role of dispatchers, the simplified algorithm for intervention and the universal encouragement to use an AED ensure that any bystander -trained or not—can initiate effective resuscitation within seconds.'*

Adult advanced life support

The 2025 ERC Guidelines redefine ALS by integrating prevention, precision treatment and quality assurance. Prevention becomes a core ALS pillar, mandating early warning systems, staff training and clear escalation pathways for in-hospital cardiac arrest, while out-of-hospital strategies focus on cardiovascular risk reduction, early recognition of prodromal symptoms and systematic evaluation and family screening for inherited cardiac disorders. Cardiac arrest treatment prioritizes immediate ALS with rapid defibrillation, timely adrenaline for non-shockable rhythms, pre-assigned and trained resuscitation teams and structured audit and debriefing for continuous quality improvement. Defibrillation strategies are refined to minimize interruptions, recommend antero-posterior pad repositioning after three failed shocks, standardize biphasic energy settings (≥150 J) and limit manual defibrillation to experienced providers. Airway management follows a stepwise, skill-based approach, favouring video laryngoscopy and mandatory waveform capnography. Drug therapy retains adrenaline as first-line, amiodarone as the preferred antiarrhythmic, selective thrombolysis for suspected pulmonary embolism and restrictive fluid use, while discouraging routine calcium, bicarbonate or corticosteroids. Monitoring is strengthened through capnography as a central tool for CPR quality and ROSC detection, selective use of intra-arrest ultrasound, mechanical CPR devices and extracorporeal CPR in specialized centres.

Professor Christian Hassager (consultant cardiologist in acute and intensive cardiac care medicine):

> *'The new Advanced Life Support and post-resuscitation sections bridge emergency action with critical care precision. The emphasis on early prevention, refined defibrillation strategy, targeted physiological goals and multimodal neuroprotection underscores a unified mission: not merely restoring circulation but safeguarding the brain and heart through every phase of recovery. It's the most clinically coherent guideline we've had to date.'*

Post-resuscitation care

Post-resuscitation care is framed as a structured continuum from ROSC to long-term recovery, with emphasis on neuroprotection, multimodal prognostication and rehabilitation. Fever prevention, physiological optimization and careful neurological assessment are emphasized. Survivorship now includes systematic follow-up, psychosocial support and family involvement, alongside integration of organ donation and evaluation of unexplained cardiac arrest.

Dr. Eva-Maria Dorsch (neurologist and psychosomatic specialist with focus on post-cardiac arrest care):

> *'Cardiac arrest survivors frequently face depression, anxiety and post-traumatic distress, as shown in our Cardiac Arrest Outpatient Care Center.[2] The 2025 Guidelines rightly elevate rehabilitation and psychosocial care to the same level of importance as acute management. Recovery is not complete when the pulse returns; it continues in the mind, the body and the reintegration of the person into life.'*

In summary, the 2025 ERC Guidelines redefine success in resuscitation medicine: survival is no longer the endpoint—restoration of function, dignity and quality of life is.

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Data availability

No quantitative data was used to generate this manuscript.

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